Definite Integration: Formula Sheet

Basics

Newton-Leibnitz: $\int_a^b f(x)dx = F(b) - F(a)$, $F' = f$, F continuous

Dummy variable: $\int_a^b f(x)dx = \int_a^b f(t)dt$

Interchange: $\int_a^b f = -\int_b^a f$, $\int_a^a f = 0$

Parts:

$$\int_a^b uv' = \left[uv\right]_a^b - \int_a^b u'v$$

Limit inside: $\lim \int_p^q f_n = \int_p^q \lim f_n$ (p, q free of n)

Inverse pair:

$$\int_a^b f + \int_{f(a)}^{f(b)} f^{-1} = bf(b) - af(a)$$

Root: $\int_a^b f = 0$, f continuous $\Rightarrow f(c) = 0$, $c \in (a, b)$

Properties

P3: $\int_a^b f = \int_a^c f + \int_c^b f$

P4: $\int_{-a}^a f = \int_0^a [f(x) + f(-x)]dx$; even $2\int_0^a f$, odd 0

P5 King:

$$\int_a^b f(x)dx = \int_a^b f(a + b - x)dx$$

King, ratio:

$$\int_a^b \frac{f(x)}{f(x) + f(a + b - x)}dx = \frac{b - a}{2}$$

King, $xf(x)$: $\int_a^b xf(x)dx = \frac{a+b}{2}\int_a^b f$ if $f(a + b - x) = f(x)$

P6 Queen:

$$\int_0^{2a} f = \int_0^a [f(x) + f(2a - x)]dx$$

Queen cases: $2\int_0^a f$ if $f(2a - x) = f(x)$; 0 if $f(2a - x) = -f(x)$

P7 Periodic: $\int_a^{a+nT} f = n\int_0^T f$, $\int_{nT}^{a+nT} f = \int_0^a f$

Reciprocal limits:

$$x = \frac{1}{t}$$

$$\frac{1}{1 + b^x} + \frac{1}{1 + b^{-x}} = 1$$

Standard values

$\int_0^{\pi/2} \sin^n x = \int_0^{\pi/2} \cos^n x$: 1, $\frac{\pi}{4}$, $\frac{2}{3}$, $\frac{3\pi}{16}$ ($n = 1..4$)

Log-sine:

$$\int_0^{\pi/2} \ln \sin x \, dx = -\frac{\pi}{2} \ln 2$$

$$\int_0^{\pi/4} \ln(1 + \tan x)dx = \frac{\pi}{8} \ln 2$$

$\int_0^n [x]dx = \frac{n(n-1)}{2}$, $\int_0^n \{x\}dx = \frac{n}{2}$

Wallis:

$$\int_0^{\pi/2} \sin^n x \cos^m x = \frac{[(n-1)(n-3)\cdots][(m-1)(m-3)\cdots]}{(m+n)(m+n-2)\cdots}K$$

Wallis K: $\frac{\pi}{2}$ if m, n both even, else 1

Leibnitz and determination

Leibnitz:

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)dt = f(h)h' - f(g)g'$$

Indefinite:

$$\frac{d}{dx}\int f(x)dx = f(x)$$

x inside:

$$\int_0^x (x - t)f(t)dt = x\int_0^x f - \int_0^x tf$$

Constant limits: $\int_a^b f(t)dt = k$ (a constant)

Limit of sum

Basic:

$$\lim_{n\to\infty} \frac{1}{n}\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right) = \int_0^1 f(x)dx$$

Interval:

$$\int_a^b f = \lim \frac{b-a}{n}\sum_{r=0}^{n-1} f\left(a + \frac{(b-a)r}{n}\right)$$

Limits: $r = 1 \to pn$ gives $0 \to p$

Product: $\ln L = \int_0^1 \ln f(x)dx$

Order ($f \uparrow$):

$$\frac{1}{n}\sum_0^{n-1} f < \int_0^1 f < \frac{1}{n}\sum_1^n f$$

Estimation and reduction

Bounds: $m(b - a) \le \int_a^b f \le M(b - a)$

Concave up:

$$(b - a)f\left(\frac{a+b}{2}\right) < \int_a^b f < \frac{(b-a)(f(a) + f(b))}{2}$$

Midpoint:

$$\int_a^b f \approx \frac{b-a}{4}\left[f(a) + 2f\left(\frac{a+b}{2}\right) + f(b)\right]$$

Comparison:

$$f_1 \le f \le f_2 \Rightarrow \int f_1 \le \int f \le \int f_2$$

Modulus:

$$\left|\int_a^b f\right| \leq \int_a^b |f|$$

$$\int_0^{\pi/2} \sin^n x : \ I_n = \frac{n-1}{n} I_{n-2}$$

$$\int_0^{\pi/4} \tan^n x : \ I_n + I_{n-2} = \frac{1}{n-1}$$

$$\int_1^e (\ln x)^n : \ I_n = e - nI_{n-1}$$